

The Impact of Device Footprint Scaling on High-Performance CMOS Logic Technology

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Abstract—We propose selective scaling of device footprint for 65 nm and beyond CMOS technologies. The benefits of selective scaling of device footprint are illustrated using an ultrathin-body fully depleted silicon-on-insulator transistor as an example. We study the effect of footprint scaling on device, circuit, and system level performance. A complete 2-D device structure is modeled for the numerical analysis. The results predict that an optimal footprint design can provide 30% smaller chip layout area, 20% faster speed, and 10% less dynamic power on overall chip performance benchmarked with a 53-bit pipelined multiplier. The variability analysis on both dc and ac characteristics indicates that the benefits of selective footprint scaling are not degraded by device variation.

Index Terms—CMOS, device footprint, device scaling, device variation, numerical simulation, selective scaling, ultrathin-body (UTB) fully depleted silicon-on-insulator (FD-SOI).

I. INTRODUCTION

AS DEVICE physical gate length is reduced to below 25-nm at/beyond 65-nm technology node, various leakage currents and device parameter variation become the most important considerations for device optimization. In fact, it can be argued that reduction of gate length below 25 nm may not offer the same advantage as short-gate devices had provided historically in terms of power and performance at the system level [1]. The major detractors are: the lack of a thin equivalent gate oxide (with low leakage current) for effective short channel effect control, the increasing contribution of the fringing parasitic capacitance to the total gate capacitance, and the rising contribution of the source/drain (S/D) resistance to the total device on-resistance. A reasonable question to ask is: if the gate length scaling stops, can we achieve performance gains through a different device scaling scenario?

The size of the active channel is only a small fraction of the total device footprint. Device footprint is mainly determined by patterning tolerances, spacer width, contact, and isolation

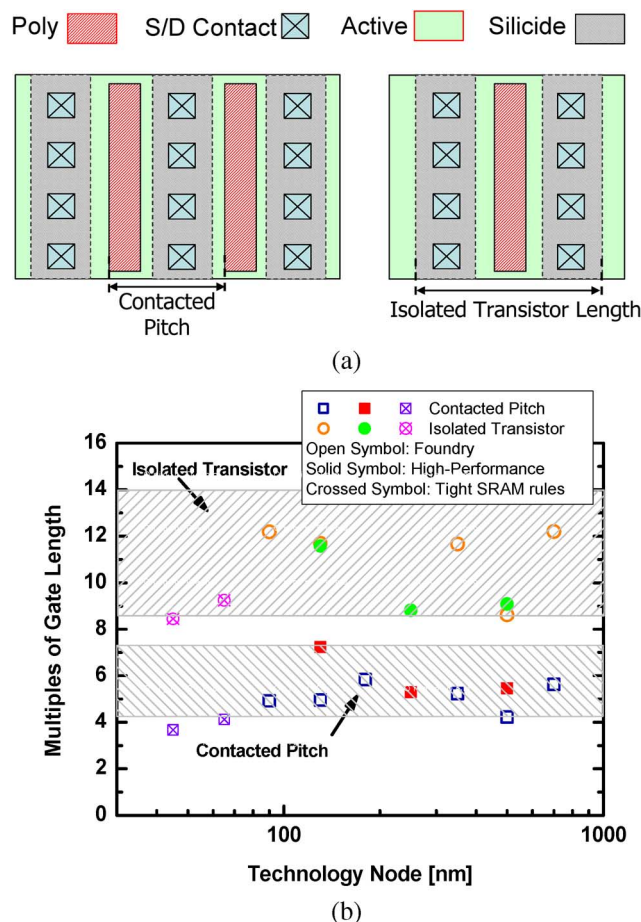


Fig. 1. (a) Definitions for “Contacted pitch” and “Isolated transistor length” transistors. (b) The total device length versus technology node. The ratio of the total device length over the gate length remains almost constant from 1 μm down to 65-nm technology node for both contacted pitch and isolated transistor. Adapted from a figure in Wong *et al.* [2] with additional data.

sizes. In fact, the scaled device footprint has stayed fairly constant throughout many generations from 1 μm to 65 nm technology. Fig. 1 shows historical data of device footprint for contacted pitch transistors and isolated transistors [2]. In this paper, we postulate that even with the gate length remaining the same, selectively scaling the device footprint will provide significant circuit-level performance improvement from technology generation to technology generation. The main reason is that reducing device footprint reduces the device parasitics and interconnect length and therefore improves speed and power efficiency. We make the bold assumption that the sizes of various device structures (e.g., contact sizes, overlay spacings)

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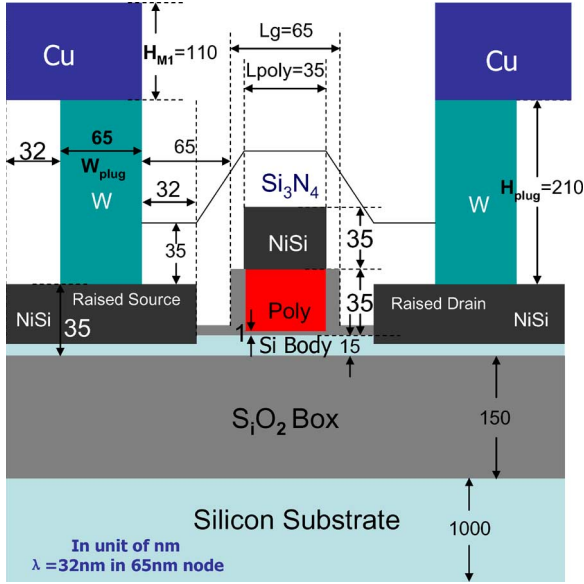


Fig. 2. Cartoon illustration of the device structure for UTB FD-SOI in 65-nm technology node, including both the gate region and the S/D contact region. All values are in unit of nanometer.

can be arbitrarily reduced using yet-to-be developed process technologies [3], [4]. The purpose of this paper is to illustrate the opportunity for a different device-scaling scenario and provide the impetus for further research in process technologies to realize this new scaling scenario.

We perform device and circuit simulations to support our postulate. The ultrathin-body (UTB) silicon-on-insulator (SOI) FET, which has a simple device structure, is used as an example device to illustrate the concept. Device density, gate delay, and power consumption are analyzed for several device footprint scaling scenarios. The performance gains are verified at the circuit-macrolevel by modeling a 53-bit multiplier using HSPICE [5] with device model parameters fitted to mixed-mode simulation results. With the same static power consumption, 30% smaller chip layout area, 20% faster speed, and 10% less dynamic power consumption can be achieved with the proposed selective footprint scaling scenario.

II. DEVICE STRUCTURE AND SCALING SCENARIOS

Device simulation and device/circuit mixed-mode simulation in this paper are performed using Taurus-Device [6]. To accurately capture the effects of device scaling on both dc and ac characteristics, a complete 2-D device structure including both the gate region and the S/D contact regions for the UTB fully depleted SOI (FD-SOI) FET is used for device simulation. A schematic cross section of the device structure with standard digital circuit design rules is illustrated in Fig. 2. The Si body thickness is 15 nm and is fully depleted. The S/D contact plugs are included in the model in order to accurately model the parasitic coupling between the gate and S/D contact region of the device. The raised S/D silicide region affects both the dc and the ac performance. The physical gate length is 35 nm. The physical models and the parameters used in the numerical simulation are chosen to best fit the simulation results with

TABLE I
SUMMARY OF DC PARAMETERS FOR THE SIMULATED DEVICE
STRUCTURE IN FIG. 2 WITH 1- μ m GATE WIDTH

	Channel Doping (cm ⁻³)	V _{ds} =0.05V			V _{ds} =1V			DIBL (mV/V)	S _{sub} (mV/dec)
		I _{off} (nA)	I _{on} (mA)	V _{th} (V)	I _{off} (nA)	I _{on} (mA)	V _{th} (V)		
nFET	5.40E+18	2.33	0.14	0.34	111	0.85	0.19	148	90
pFET	5.25E+18	1.78	0.06	-0.39	107	0.55	-0.21	179	96

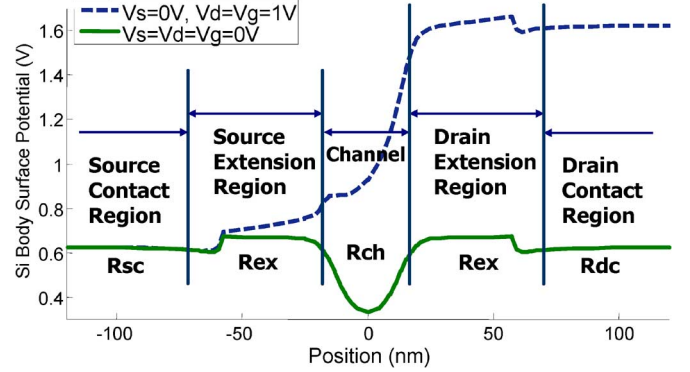


Fig. 3. Silicon body surface potential profile from source contact to drain contact. The lower curve is for OFF-state ($V_s = V_d = V_g = 0$ V), and the upper curve is for the ON-state ($V_s = 0$ V, $V_d = V_g = 1$ V). Significant potential drops over the S/D extension region.

typical 65-nm node experimental data. The effects of stress-enhanced mobility on device characteristics [7] are not treated in this paper in order to clearly delineate the interplay between device parasitics and circuit performance. It is expected that the loss of stress-dependent mobility enhancement as a result of smaller S/D area will lower the overall performance. However, the general trend illustrated in this paper is still applicable. The major dc characteristics of the device with standard digital circuit design rules are summarized in Table I.

Both the ON-current and the gate capacitance determine the device/circuit dynamic performance. For sub-100 nm CMOS devices, there are significant potential drops over the S/D extension region and contact region (Fig. 3). The ratio of the parasitic resistance (R_{par}) (including S/D extension resistances R_{ex} and contact spreading resistance R_c) to the total channel resistance for FD-SOI becomes even larger due to the UTB film and the lack of the S/D deep junction. An effective way to reduce R_{par} is to use silicided raised source and drain at the cost of increased parasitic coupling capacitance between the gate and the raised S/D region. An alternative approach is to reduce the distance L_{sd} between the gate edge and S/D contact stud. This approach also results in increased gate to S/D coupling capacitance.

We examine four device footprint scaling scenarios (Fig. 4). The first scaling scenario “s1” only reduces the distance L_{sd} to make the total device length smaller. The second scenario “s2” scales down the dimensions of the S/D contact plug (stud) by half. In the third scenario “s3,” the nitride spacer in gate region is replaced with a low- κ dielectric (assumed to have the dielectric constant of silicon dioxide), and all the dimensions of the S/D region are scaled down by half. Finally, in the fourth scenario “s4,” the raised S/D is not implemented to eliminate the parasitic coupling between the gate and raised S/D. In the following analysis, both dc and ac characteristics

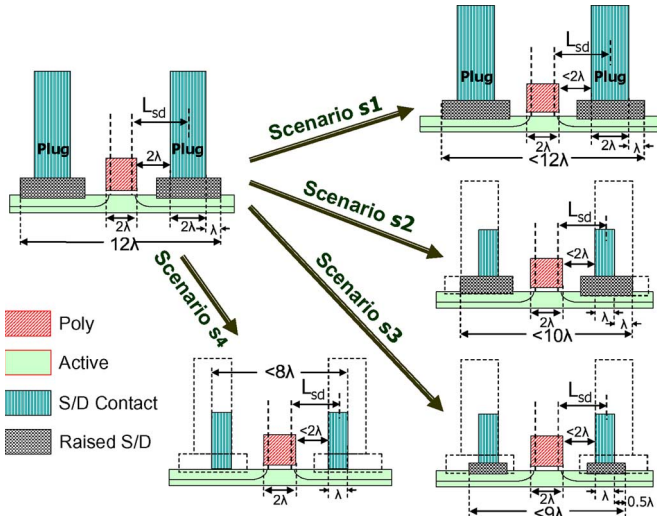


Fig. 4. Device cross section of devices with different footprint scaling scenarios. Scenario "s1": only reduces the distance L_{sd} to make the total device length smaller. Scenario "s2": scaling down the dimensions of S/D contact plug (stud) by half. Scenario "s3": the nitride spacer in gate region was replaced with silicon dioxide, and all the dimensions of the S/D region were scaled down by half. Scenario "s4": the raised S/D region was removed in order to eliminate the parasitic coupling between the gate and raised S/D region.

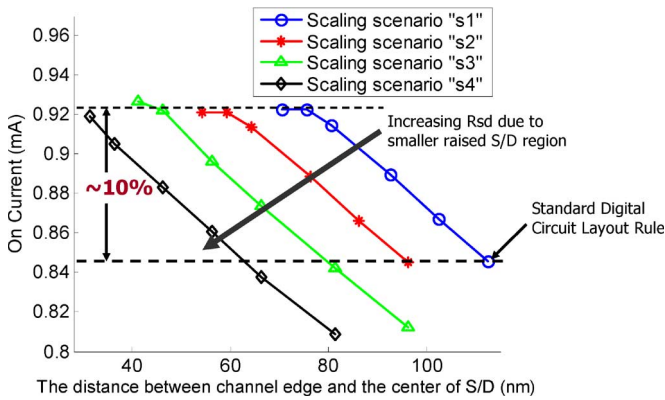


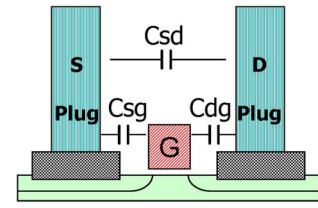
Fig. 5. Device saturation ON-current versus the distance L_{sd} between the gate edge and S/D, for four scaling scenarios. The ON-current first increases with reduced L_{sd} , then saturates. Larger dimension of raised S/D region results in higher ON-current due to smaller R_{par} .

are evaluated as a function of L_{sd} for all four scaling scenarios. The OFF-current for all the cases studied are set to the same value by setting the appropriate doping concentration in the channel region.

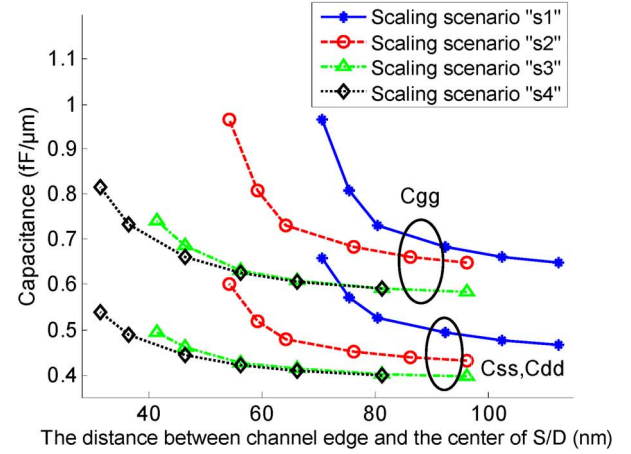
III. DEVICE AND SIMPLE CIRCUIT PERFORMANCE ANALYSIS

First, we analyze the behavior of the saturation ON-current and the device capacitances as a function of the distance (L_{sd}) between the gate edge and the S/D contact stud for various scaling scenarios described above.

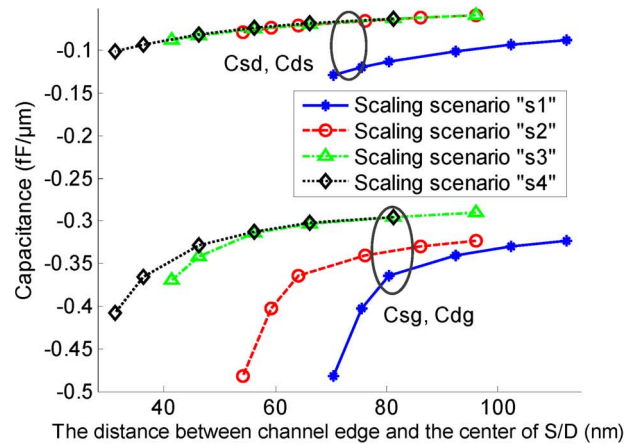
The device ON-current increases with reduced distance L_{sd} between the gate edge and S/D contact stud mostly due to the smaller S/D extension resistance R_{ex} (Fig. 5). The S/D contact spreading resistance has little dependence on the footprint scaling scenarios given the fact that the current transfer length



(a)



(b)



(c)

Fig. 6. (a) C_{sg}/C_{dg} is the coupling capacitance between S/D and gate region, C_{sd} is the coupling capacitance between source and drain. (b) The gate/S/D node capacitance, and (c) $C_{sg}/C_{dg}/C_{sd}$ transcapacitance as a function of L_{sd} for four scaling scenarios. The capacitances increase with reduced L_{sd} . Larger dimension of raised S/D region results in higher parasitic coupling among gate, source, and drain.

is shorter than the contact length in the cases studied. The maximum ON-current improvement of the scaled device over the device with the standard digital circuit design rule (termed as "normal device" in the following) is about 10%.

When evaluating dynamic performance (circuit speed), capacitance effects are also significant. As shown in Fig. 6, the gate/S/D node capacitances increase with reduced L_{sd} [Fig. 6(b)] mostly due to the increased coupling capacitance C_{sg}/C_{dg} [Fig. 6(a)] between gate region and S/D contact region [Fig. 6(c)]. The above observation also indicates that it is possible to further improve the circuit dynamic performance by reducing the coupling capacitance C_{sg}/C_{dg} with a smaller gate

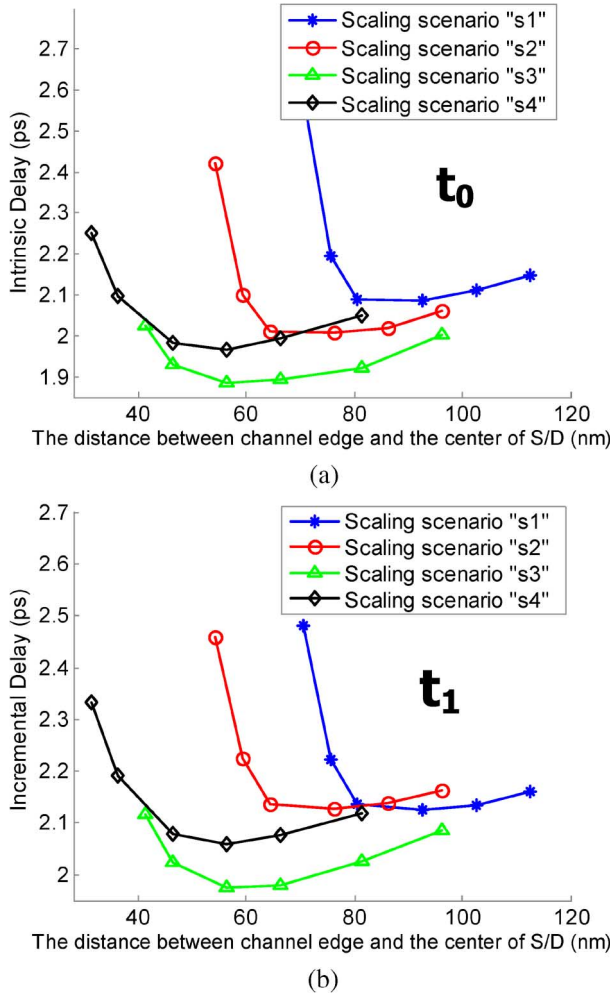


Fig. 7. Both (a) the inverter intrinsic delay (without load) t_0 , and (b) the inverter incremental delay per FO load (t_1) first decrease slowly, then increase quickly with reduced L_{sd} . Scenario "s3" (with half-sized S/D dimensions) has the minimum delay among the four scaling scenarios studied.

height. The larger dimensions of the raised S/D region give both higher ON-current (Fig. 5) and larger capacitances (Fig. 6).

A three-stage ring-oscillator (RO) and a four-stage inverter chain are used to examine the dynamic performance. Circuit simulations are performed using device/circuit mixed-mode numerical simulation in Taurus-Device [6]. The device width ratio between pFET and nFET is designed at 1.5 to balance the pull-up and the pull-down delay. The inverter delay (τ) is evaluated between the 50% to 50% points, and can be expressed as a linear function of the fan-out (FO) load $\tau = t_0 + t_1 \cdot FO$, where t_0 is the circuit intrinsic delay without any load, and t_1 is the incremental delay per FO load. As shown in Fig. 7, the values of t_0 and t_1 are almost the same because the value of the intrinsic gate capacitance is close to the sum of the drain-to-gate and source-to-gate coupling capacitances for SOI device with negligible coupling between gate/S/D and substrate. Both t_0 and t_1 follow the same trend, first decreasing slowly then going up quickly with reduced L_{sd} because of the competition between the increasing saturating ON-current and increasing capacitances. As a result, the total inverter delay follows a similar trend (Fig. 8). An optimal point for minimum delay exists for each scaling scenario. Although the device without

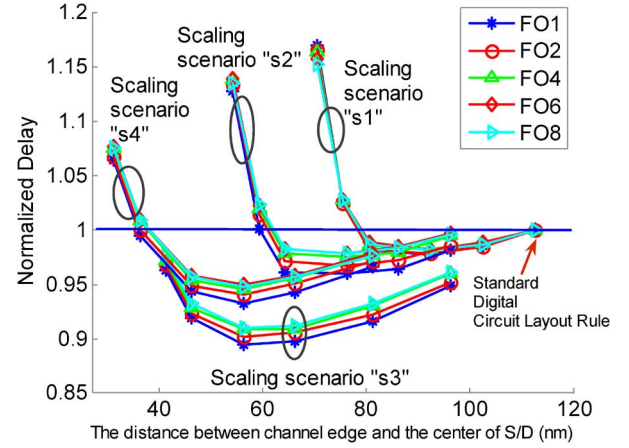


Fig. 8. Inverter delay first decreases slowly, then increases quickly with reduced L_{sd} . An optimal design point with minimum delay exists for each scaling scenario. The maximum speed improvement of the scaled device over the normal device is about 10% for a variety of FO load.

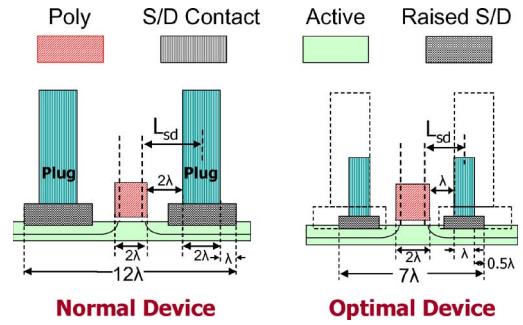


Fig. 9. Device cross-sectional comparison between the device with standard digital circuit design rule (normal device) and the device with selectively scaled S/D footprint (optimal device).

raised S/D region in the fourth scaling scenario gives the smallest capacitance (Fig. 6), the degraded drive current (Fig. 5) slows down the circuit (Figs. 7 and 8). A tradeoff between the ON-current and capacitances exists when sizing the dimensions of S/D region. Fig. 8 also indicates that the most effective way to improve the device speed for UTB FD-SOI is reducing the size of the raised S/D region whose effect is denoted by the difference between scenario "s2" and scenario "s3." Reducing the contact plug size (scaling scenario "s2") helps reduce the total device length. For the cases studied, the device with scaling scenario "s3" at the point L_{sd} around 55 nm has about 10% faster inverter speed than the normal device. At the optimal design point, the ON-current improvement over normal device is about 7%, and the total device length is about 6.6λ which is 45% smaller (isolated device) layout area than the normal device. At $L_{sd} = L_{sd0}$ (6.6λ) the delay is at minimum. In the following section, we use an optimal value of $L_{sd} = 7\lambda$ for the following reasons: 1) the delay is sensitive to L_{sd} variation with L_{sd} less than L_{sd0} , but is rather flat in the region with L_{sd} slightly larger than L_{sd0} ; 2) it is more convenient to set up the design rule to be a multiple number of λ ; 3) a relaxed design with smaller parasitic capacitance helps to reduce dynamic power consumption with very minor speed/area penalty. The cross-sectional comparison between the normal device and the optimal device is summarized in Fig. 9.

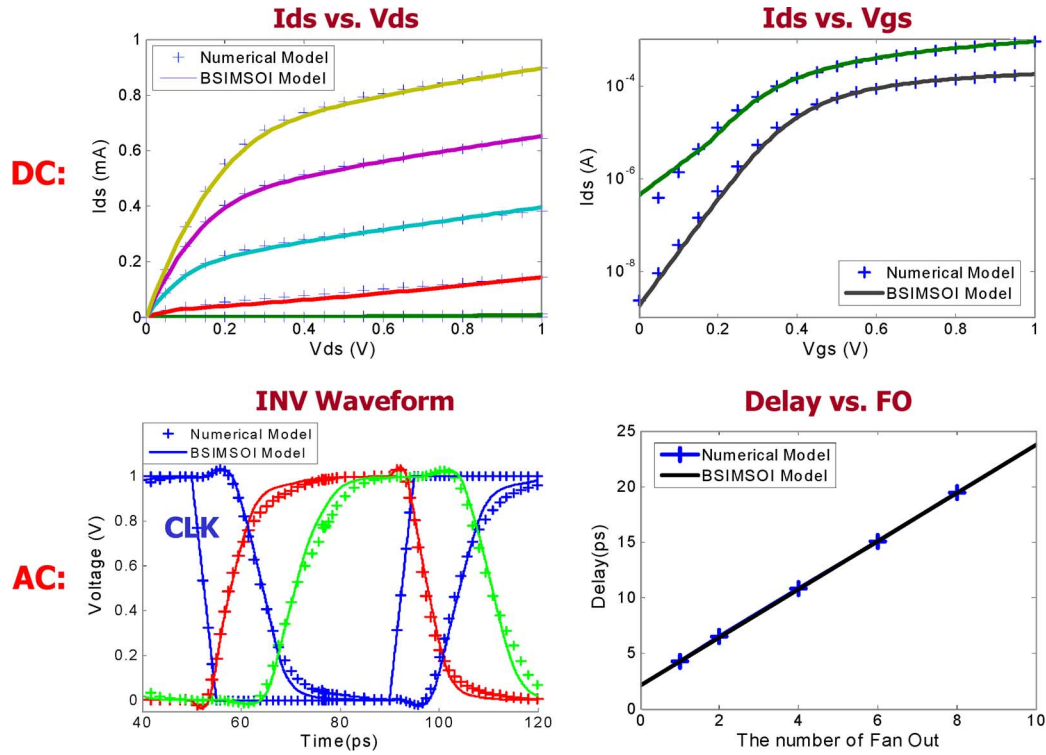


Fig. 10. Simulation results comparison between the numeric device model and extracted BSIM3vSOI compact model in both dc (I_{ds} versus V_{ds} , and I_{ds} versus V_{gs}) and ac (waveform, and delay versus FO) characteristics.

IV. SYSTEM LEVEL PERFORMANCE ANALYSIS

The previous section presented results for small circuits with very little wiring load. For larger circuit structures, the interconnect delay is comparable to the gate delay. The smaller device dimension permits the use of shorter interconnects, hence results in higher speed and lower power consumption at the system level. A fully custom designed 53-bit multiplier using a seven-metal layer technology is used to evaluate the benefits of device footprint scaling on complex circuits. Since it is impossible to perform mixed-mode simulation for such a large circuit, we first extract device parameters from the Taurus-Device simulator into the BSIM3vSOI compact model using Aurora [9]. Fig. 10 shows good agreement between the numerical device-simulation results (Taurus) and the extracted BSIM3vSOI compact model for both dc and ac characteristics.

Pipelined dual-rail dynamic domino logic is chosen for the circuit design. We use the Booth algorithm to calculate the partial product, and all the partial products are summed up with a three-stage pipelined Wallace tree (Fig. 11). The latency time is two clock cycles and the throughput is one output per cycle. With the folded chip floor plan, there are two critical paths [Fig. 11(b)] with long routing interconnects: 1) [solid arced arrow in Fig. 11(b)] a long vertical wire (segment A) and a long horizontal wire (segment B) between the outputs of the Booth encoder and the inputs of the partial-product generators in the first stage and 2) [dashed arced arrow in Fig. 11(b)] a long vertical wire (segment C) between the second level and the third level Wallace tree within the second stage, and a long vertical wire (segment D) between the latch outputs of the second stage and the inputs of the fourth level Wallace

tree in the third stage. All the interconnect capacitance (about $0.2 \text{ fF}/\mu\text{m}$) and resistance extracted from the chip floor plan and the routing map are included in the SPICE simulation. A minimum positive setup time of 15 ps for latches is guaranteed in evaluating the maximum operation frequency. The circuit layout area can be expressed as the summation of the device layout area and the routing area, where single device layout area is given by $\alpha \cdot \text{Area}_{\text{gate}}$. The parameter α is the ratio of the total device length to the gate length (6 for the normal device and 3.5 for the optimal device). The routing areas for the circuits built with both the normal device and the optimal device are approximately the same with the same interconnect technology and the same design rule for the metal layers (i.e., the same minimum width and minimum pitch for metal layers).

Fig. 12 shows the performance comparison between the multiplier using the normal device and the optimal device with a 1-V power supply. Compared with the multiplier built with the normal device, the multiplier built with the optimal device occupies 30% less layout area, operates at 20% higher speed (10% is due to the faster device, another 10% comes from the shorter interconnects), and consumes 10% less dynamic power with the same operation frequency. The dynamic power reduction is mostly due to the smaller interconnect capacitance of the smaller circuit layout area.

V. VARIABILITY ANALYSIS

One major reason that prohibits the further device scaling is the performance variation caused by the increased process variations beyond 65-nm technology node. Thus, it is important

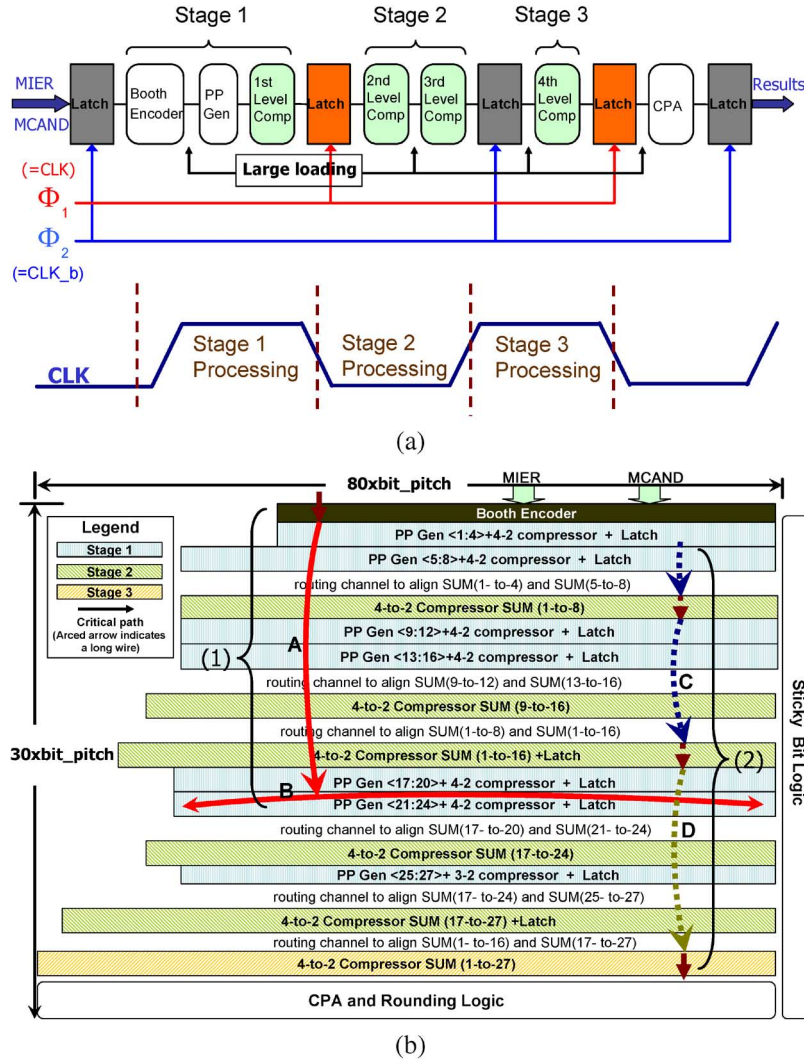


Fig. 11. (a) Logic architecture, and (b) the illustration of chip floor plan for the fully custom designed 53-bit pipelined multiplier. There are two critical paths with long routing interconnects: 1) a long vertical wire (segment A) and a long horizontal wire (segment B) between the outputs of the Booth encoder and the inputs of the partial-product generators in the first stage and 2) a long vertical wire (segment C) between the second level and the third level Wallace tree within the second stage, and a long vertical wire (segment D) between the latch outputs of the second stage and the inputs of the fourth level Wallace tree in the third stage.

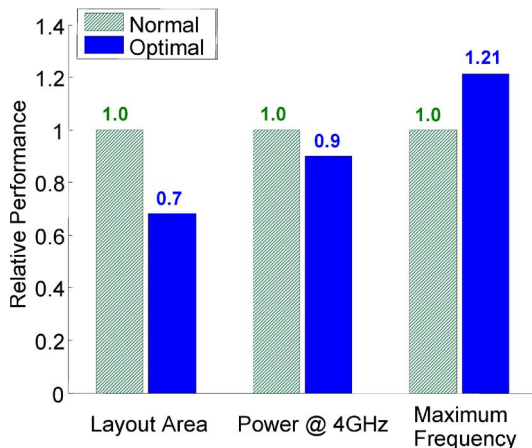


Fig. 12. Performance comparison between the multiplier made with optimal device (solid bars) and the multiplier made with normal device (shaded bars). With the optimal device, the multiplier achieves 30% smaller die area, 20% higher operation frequency, and 10% less dynamic power consumption.

to investigate the performance variability for the optimal device. We assumed a superhalo profile [8], and the number of dopants in channel region was assumed to be constant. We use Monte Carlo simulation to study the effects of three major process related variations, gate length (L_{poly}) variation, body thickness (T_{body}) variation, and contact size (both the contact length L_{cnt} and the height of raised S/D region H_{cnt}) variation, on both dc (V_{th}) and ac (FO1 delay) characteristics. These variations are treated as independent processes. The parameters and simulation results are summarized in Table II. Although the footprint scaled device is a little more sensitive ($\sim 0.2\%$) to the contact size variation because of the smaller contact size than the normal device, the effect of contact size variation is secondary compared to the effects of L_{poly} variation and T_{body} variation (Table II). The scaled footprints of the optimal device do not increase the performance variation in terms of both dc and ac characteristics (Table II). The average values are close to the nominal values. FO1 delay is less sensitive to T_{body}

TABLE II
 V_{th} AND FO1 DELAY VARIABILITY ANALYSIS FOR BOTH NORMAL
 DEVICE AND OPTIMAL DEVICE WITH GATE LENGTH (L_{poly})
 VARIATION, BODY THICKNESS (T_{body}) VARIATION, AND CONTACT SIZE
 (BOTH CONTACT LENGTH L_{cnt} AND THE HEIGHT OF RAISED S/D
 REGION H_{cnt}) VARIATION (ASSUMING 10%/3 σ VARIATION).
 BOTH L_{cnt} AND H_{cnt} SCALE IN THE SAME DIRECTION TO
 REPRESENT THE WORSE CASE FOR CONTACT SIZE VARIATION

		Vth (mV)			FO1 Delay (ps)		
		Nominal Value	Mean Value	1 σ Variation (% of mean value)	Nominal Value	Mean Value	1 σ Variation (% of mean value)
Normal Device	L_{poly}	174.2	173.7	4.7 (2.71%)	4.10	4.11	0.093 (2.26%)
	T_{body}	174.2	173.8	11.3 (6.50%)	4.10	4.13	0.087 (2.10%)
	L_{cnt}/H_{cnt}	174.2	174.2	< 0.1 (0.06%)	4.10	4.12	0.025 (0.61%)
	Total	174.2	173.9	12.24 (7.04%)	4.10	4.12	0.130 (3.15%)
Optimal Device	L_{poly}	175.1	174.6	4.7 (2.69%)	3.70	3.68	0.097 (2.64%)
	T_{body}	175.1	174.8	11.4 (6.52%)	3.70	3.71	0.072 (1.93%)
	L_{cnt}/H_{cnt}	175.1	176.0	1.2 (0.7%)	3.70	3.69	0.031 (0.85%)
	Total	175.1	175.1	12.39 (7.08%)	3.70	3.69	0.125 (3.38%)

variations than V_{th} . Using a first-order approximation, the delay due to the inner fringing capacitance can be expressed as

$$\tau_{fr} \propto R_{on} \cdot C_{gsd,fr} \propto \frac{1}{T_{body}} \cdot T_{body} \sim \text{Constant}.$$

The weak dependence of FO1 delay on body thickness is due in part to the cancellation of the gate to S/D inner channel fringing capacitance of the load devices and the on resistance of the driving device for FD-SOI.

VI. SUMMARY AND DISCUSSIONS

In this paper, we propose a new device scaling scenario for sub-65 nm technology node high-performance CMOS technology. We postulate that even with the gate length remaining the same, selectively scaling the device footprint will provide significant circuit-level performance improvement from technology generation to technology generation. For small benchmark circuits such as ring oscillators and inverter chains, a selectively scaled device with reduced footprint achieves 45% smaller device layout area, 7% higher drive current, and 10% shorter delay than the normal (standard) device. For a fully custom designed 53-bit multiplier, the selectively scaled device with reduced footprint occupies 30% less layout area, operates at 20% higher frequency, and consumes 10% less dynamic power while keeping the static power dissipation the same.

Ever since the 0.35- μm node, the gate length has been selectively scaled to improve performance. Gate length selective scaling has been effective when the gate delay of unloaded (or lightly loaded) circuits were dominated by the intrinsic gate capacitance. Beyond the 65-nm technology node, the physical gate length is less than 1/10 of that at 0.35 μm node, but the parasitic gate to S/D capacitance/resistance remains roughly the same. Therefore, further aggressive gate length scaling does not bring the same benefits as it did when the gate lengths were longer. At the same time, interconnect delay becomes a larger fraction of the total circuit delay. Therefore, a new selective scaling scenario where the parasitic capacitance/resistance and the interconnect delay are minimized is effective. The analyses of this paper suggest that this new selective scaling scenario can achieve significant performance improvement.

Innovative process technologies such as self-assembly [3], [4] and low- κ spacer will help realize the selective footprint scaling strategy. Additionally, packaging and/or architectural solutions are also required to mitigate the increased power density due to the smaller device footprint. Future work on this subject may further consider the effects of stress-dependent carrier mobility [7], 3-D modeling of contact stud capacitance, more accurate modeling of the S/D resistance including contact resistance for future technology nodes, the use of a smaller gate height (use of metal gate), the use of low- κ spacer and low- κ isolation dielectric (low- κ STI), simultaneous optimization of footprint, and device width and the case of bulk CMOS where the S/D junction capacitance plays a more important role than UTB SOI FET.

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Ching-Te Chuang (S'78–M'82–SM'91–F'94) received the B.S.E.E. degree from National Taiwan University, Taipei, Taiwan, R.O.C., in 1975, and the Ph.D. degree in electrical engineering from University of California, Berkeley, in 1982.

From 1977 to 1982, he was a Research Assistant in the Electronics Research Laboratory, University of California, Berkeley, working on bulk and surface acoustic wave devices. He was with the IBM T.J. Watson Research Center, Yorktown Heights, NY, in 1982, working on scaled bipolar devices, technology, and circuits.

He studied the scaling properties of epitaxial Schottky barrier diodes, did pioneering works on the perimeter effects of advanced double-poly self-aligned bipolar transistors, and designed the first subnanosecond 5-kb bipolar emitter-coupled logic static random access memory (SRAM). From 1986 to 1988, he was Manager of the Bipolar VLSI Design Group, working on low-power bipolar circuits, high-speed high-density bipolar SRAMs, multigigabits per second fiber-optic data-link circuits, and scaling issues for bipolar/BiCMOS devices and circuits. Since 1988, he has been managing the High Performance Circuit Group, investigating high-performance logic, and memory circuits. Since 1993, his group has been primarily responsible for the circuit design of IBM's high-performance CMOS microprocessors for enterprise servers, PowerPC workstations, and game/media processors. Since 1996, he has been leading the efforts in evaluating and exploring scaled/emerging technologies, such as partially depleted/SOI, ultrathin/SOI, strained-Si devices, hybrid orientation technology, and multigate/FinFET devices, for high-performance logic and SRAM applications. He is the holder of 18 U.S. patents with another 18 pending. He has authored or coauthored over 240 papers.

Dr. Chuang served on the Device Technology Program Committee for IEEE International Electron Devices Meeting (IEDM) in 1986 and 1987, and the Program Committee for Symposium on VLSI Circuits from 1992 to 2006. He was the Publication/Publicity Chairman for Symposium on VLSI Technology and Symposium on VLSI Circuits in 1993 and 1994, and the Best Student Paper Award Sub-Committee Chairman for Symposium on VLSI Circuits from 2004 to 2006. He was elected an IEEE Fellow in 1994 "For contributions to high-performance bipolar devices, circuits, and technology." He has authored many invited papers in international journals such as *International Journal of High Speed Electronics*, *PROCEEDINGS OF THE IEEE*, and *IEEE Circuits and Devices Magazine*. He has presented numerous plenary, invited or tutorial papers/talks at international conferences such as International SOI Conference, Design Automation Conference (DAC), VLSI Technology, Systems, and Applications, International Solid State Circuits Conference (ISSCC) Microprocessor Design Workshop, VLSI Circuit Symposium Short Course, IEEE International Symposium on Quality Electronic Design, International Conference on Computer-Aided Design, Asia-Pacific Microwave Conference, VLSI Design, Automation, and Test, and IEEE International Symposium on Circuits and Systems, etc. He has received one Outstanding Technical Achievement Award, one Research Division Outstanding Contribution Award, five Research Division Awards, and ten Invention Achievement Awards from IBM. He was the corecipient of the Best Paper Award at the 2000 IEEE International SOI Conference.



H.-S. Philip Wong (S'81–M'82–SM'95–F'01) received the B.Sc. degree (Hons.) from University of Hong Kong, Kowloon, Hong Kong, in 1982, the M.S. degree from State University of New York at Stony Brook, in 1983, and the Ph.D. degree from Lehigh University, Bethlehem, PA, all in electrical engineering.

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Dr. Wong serves on the IEEE Electron Devices Society (EDS) as elected AdCom member. He serves on the IEDM committee from 1998 to 2006 and is the Technical Program Chair in 2006. He served on the ISSCC program committee from 1998–2004, and was the Chair of the Image Sensors, Displays, and microelectromechanical systems subcommittee from 2003–2004. He is the Editor-in-Chief of the IEEE TRANSACTIONS ON NANOTECHNOLOGY. He is a Distinguished Lecturer of the IEEE EDS and Solid-State Circuit Society. He has taught several short courses at the IEDM, ISSCC, Symposium on VLSI Technology, SOI conference, European Solid-State Device Research Conference, and The International Society for Optical Engineers conferences. He is a member of the Emerging Research Devices Working Group of the International Technology Roadmap for Semiconductors.